UX INTERACTION DESIGN & SCENARIO SPECIFICATION

ENVISIONED DESIGN SCENARIOS FOR COURSE REGISTRATION

Course: UX Design: From Concept to Prototype (University of Michigan) | **Evaluation:** Peer-Review Assignment

Design Context & Scenario Rationale: Building directly upon the 20 initial ideation sketches (Assignment 1) and the two diverse user personas (Minh 'Leo' Nguyen and Linh An Tran), the following two scenarios demonstrate the envisioned functionality of our collaborative course registration and timetable management solution. Each scenario satisfies the five fundamental components defined in lecture: **Setting, Agents, Goals, Actions, and Events**.

SCENARIO 1: Synchronous Team Timetable Coordination & 1-Touch Batch Enrollment

The Five Key Scenario Components:

- **Setting:** Wednesday morning at 8:50 AM (10 minutes before the registration window opens). Minh is at his desk in his off-campus apartment with a dual-monitor workstation, connected to high-speed home internet and an active Discord voice channel with his capstone group.
- **Agents:** Minh 'Leo' Nguyen (Lead User / Persona 1) and his 3 software engineering capstone teammates (Khoa, Duy, and Hoang).
- **Goals:** Coordinate schedules so all 4 members enroll in the identical practical lab section (PRN211 - Section 02), keep Tuesdays and Thursdays completely open for industry internship work, and successfully submit all 15 semester credits in under 60 seconds without server lag collisions.
- **Actions:** Minh opens the collaborative 'Group Sandbox' module (Sketch 7); invites his 3 teammates via link; watches their draft schedules populate live onto a shared weekly calendar grid (Sketch 6); uses the conflict resolver wizard (Sketch 4) to shift an overlapping elective to Wednesday afternoon; stages all approved classes into the floating tray (Sketch 3); and clicks the pre-loaded '1-Touch Instant Enroll' button (Sketch 11) the split-second the clock strikes 9:00:00 AM.
- **Events:** The registration server receives intense traffic spikes at 9:00 AM. The system recognizes the pre-validated group payload, grants a 10-minute slot reservation hold, and renders an animated progress state. A green confirmation banner (Sketch 15) announces all 4 members are successfully registered in Section 02. Google Calendar automatically syncs the verified schedule (Sketch 18).

Detailed Scenario Narrative:

It is 8:52 AM on registration morning. Minh sits at his dual-monitor setup with his class Discord voice call active. Instead of coordinating frantically via messy spreadsheets, Minh opens the portal's new *Group Sandbox* tab. He sends an invite link to Khoa, Duy, and Hoang. As each teammate joins, their proposed weekly blocks appear in distinct colors on the shared timetable grid. The system immediately flags a bright red collision: Khoa has a general elective overlapping with PRN211 Section 02 on Thursday morning. Minh clicks the *Conflict Resolution Wizard*, which highlights an alternative Section 04 for Khoa on Wednesday afternoon. Khoa approves the swap in one click. With zero timetable conflicts remaining and all 4 teammates synchronized, Minh and his team stage their 5 respective classes into their persistent floating bottom trays. The large countdown timer displays *00:00:03*. At exactly 9:00:00 AM, the central button turns bright green, reading *'1-Touch Instant Enroll'*. Minh clicks it instantly. Despite 3,000 students hitting the university servers simultaneously, the system's pre-load buffer processes the request seamlessly. Within 4 seconds, a celebratory green confirmation banner appears: *'All 5 Courses Confirmed! Team Section PRN211-02 Locked.'* Minh breathes a sigh of relief as his confirmed schedule instantly auto-syncs with his Google Calendar, leaving his Tuesdays and Thursdays completely free for work.

Relevant Sketches Envisioned: Sketch 3 (Floating Staging Tray), Sketch 4 (Conflict Wizard), Sketch 6 (Split-Screen Calendar), Sketch 7 (Multi-Scenario Sandbox), Sketch 11 (1-Touch Enroll), Sketch 15 (Confirmation State), Sketch 18 (Calendar Sync).

SCENARIO 2: Mobile In-Transit Course Swapping & Asynchronous Peer Handoff

The Five Key Scenario Components:

- **Setting:** Thursday morning at 7:40 AM. Linh An is standing on a crowded public bus (Route 27) commuting to campus, holding her iPhone in one hand with intermittent 4G mobile data and wearing headphones.
- **Agents:** Linh An Tran (Lead User / Persona 2) and an automated system waitlist bot, with downstream collaboration with her study partner Mai.
- **Goals:** Swap an inconvenient afternoon lecture in 'International Trade Law' (which directly conflicts with her mandatory 1:00 PM retail cashier shift) into a newly opened 8:00 AM morning section, while ensuring she has enough walking time between campus buildings and sharing course material folders with Mai.
- **Actions:** Linh An receives a mobile push notification (Sketch 9); taps the alert to authenticate via FaceID; reviews the side-by-side swap dialog showing instructor rating and campus walking distance (Sketches 12 & 13); notices the 8:00 AM class is in Alpha Hall, leaving a safe 20-minute walk to her 10:00 AM class; taps 'Confirm Instant Swap'; and sends a shared lecture-notes folder link to Mai asynchronously.
- **Events:** A student unexpectedly drops the 8:00 AM class, triggering the automated notification alert. While Linh An is confirming the swap, the bus enters a tunnel, briefly dropping cellular connection. The portal's offline-resilient microservice holds the slot reservation for 3 minutes (Sketch 19). Upon exiting the tunnel, the transaction finalizes successfully with a confirmation badge.

Detailed Scenario Narrative:

Linh An is commuting to university on a crowded morning bus. She had been forced to register for an afternoon lecture in International Trade Law (Section 03, 1:30 PM) because morning sections were filled. However, this afternoon slot directly conflicts with her 1:00 PM retail work shift, putting her job at immediate risk. She had previously enabled a *'Smart Waitlist Alert'* for any morning opening. At 7:42 AM, her phone buzzes with a high-priority push notification: ■ *Seat Opened: Trade Law Section 01 (Tue/Thu 08:00 - 10:00 AM). Tap to claim in 3 mins!'* Linh An taps the banner immediately. The mobile interface opens directly to a focused *Quick-Swap Dialog*. The screen highlights two vital details: (1) Switching will eliminate her work conflict completely; and (2) The automated walking calculator warns that moving from Alpha Hall at 10:00 AM to Gamma Hall at 10:15 AM requires a 12-minute walk across campus—tight, but feasible. Just as she taps *'Confirm Instant Swap'*, the bus enters a highway tunnel and cellular connection drops. Thanks to the system's 3-minute transaction hold and offline resilience mascot, her reservation does not expire. As the bus emerges from the tunnel, the screen refreshes to a green success state: *'Swap Finalized! Your afternoon is now free.'* Linh An taps the share icon to send the new section syllabus and asynchronous shared notes folder to her classmate Mai on Messenger, knowing her job and graduation schedule are both secure.

Relevant Sketches Envisioned: Sketch 2 (Seat Availability), Sketch 9 (Mobile Push Alert), Sketch 10 (Mobile Use in Transit), Sketch 12 (Section Comparator), Sketch 13 (Walking Distance Calculator), Sketch 17 (Waitlist Tracker), Sketch 19 (Server Lag Resilience).

FIVE CORE SCENARIO COMPONENTS CROSS-REFERENCE MATRIX

Component	Scenario 1: Minh (Synchronous Team Sync)	Scenario 2: Linh An (Mobile In-Transit Swap)
1. Setting	Apartment desk, dual monitors, Discord call, 8:50 AM registration launch.	Crowded public bus on Route 27, iPhone 4G cellular, 7:40 AM transit.
2. Agents	Minh (Lead Capstone Coordinator) + 3 teammates (Khoa, Duy, Hoang).	Linh An (Commuter student) + Automated waitlist bot + Study partner Mai.
3. Goals	Enroll entire 4-person team into same lab section; keep Tue/Thu workdays.	Swap conflicting afternoon class to morning to protect retail work shift.
4. Actions	Invite teammates to sandbox, resolve conflict, stage classes, 1-touch enroll.	Tap push notification, verify walking distance, confirm instant swap on phone.
5. Events	3,000-user server traffic surge; pre-validated payload executes in 4 seconds.	Seat drops unexpectedly; 3-min reservation hold survives tunnel network loss.